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RESEARCH-ARTICLE

Moroccan legal assistant Enhanced by Retrieval-Augmented Generation Technology

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ABSTRACT

This paper introduces a novel approach and its experimentation in the field of legal tech, utilizing the Retrieval Augmented Generation (RAG) model. Unlike traditional approaches, RAG enables the chatbot to provide accurate and contextually relevant legal information by accessing a vast external knowledge base. This choice overcomes the limitations of standard fine-tuning methods in AI, enhancing the chatbot's ability to offer dynamic and up-to-date legal assistance.

In this paper, we provide a detailed account of the development process and the integration of RAG into our legal tech solution. Additionally, we evaluate the performance of our approach, placing particular emphasis on its effectiveness in simplifying legal consultations. By leveraging RAG, our project showcases the potential of this technology in the field of legal tech, representing a significant step towards accessible and efficient AI-driven legal assistance.

Through our experimentation and analysis, we demonstrate the transformative impact of RAG in enabling chatbots to provide accurate and contextually relevant legal information. This advancement holds great promise for making legal assistance more accessible and efficient, paving the way for a new era of AI-driven legal tech solutions.

CCS CONCEPTS

• **Retrieval-Augmented Generation, fine-tuning, Natural Language Processing;**

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1 INTRODUCTION

Moroccan law is characterized by unique legal statutes and frequent updates, posing challenges for legal professionals and the general public seeking to comprehend and navigate its complexities. Our

chatbot aims to simplify this process by providing instant, accurate, and user-friendly legal assistance. By incorporating the RAG method, the chatbot surpasses traditional information retrieval methods, offering nuanced and contextually relevant legal advice. This not only enhances the efficiency of legal consultations but also makes legal knowledge more accessible to a broader audience in Morocco.

In the first part of this paper, we will present a comprehensive overview of five different works in the field, highlighting the current state-of-the-art. Subsequently, we will delve into our own approach and the methodologies we employed to optimize our planning, utilizing heuristic algorithms, machine learning algorithms, and mathematical formulas. We will present our results and conduct a thorough comparison with the previously discussed works, providing a comprehensive validation process to assess the performance and reliability of our developed system. Additionally, we will explain the various features implemented in the frontend application, focusing on creating an interactive and user-friendly experience.

The primary objective of this article is to provide a detailed account of the development journey of our labor law chatbot, showcasing the innovative use of the RAG method. We will explore the technical complexities involved in building a chatbot tailored to Moroccan law, discuss the advantages of utilizing the RAG model compared to conventional approaches in AI development, and illustrate the potential impact of this technology on legal assistance in Morocco. Through this exposition, we aim to contribute to the burgeoning field of AI in legal technology and demonstrate the practical utility of AI in enhancing legal services within specific regional contexts.

2 RELATED WORK

2.1 Moroccan Law as an Application Domain

Moroccan labor law, as defined in the Labor Code [3], encompasses a wide range of regulations that govern employer-employee relationships. This comprehensive legal framework covers various aspects, including employment contracts, working hours, wage structures, occupational health and safety, and conflict resolution. The complexity of these laws poses a significant challenge for individuals without specialized legal knowledge, highlighting the importance of tools that can simplify and clarify this legal domain for both laypeople and professionals.

The emergence of chatbots represents a significant advancement in various industries, particularly in the legal field. These AI-powered assistants provide round-the-clock support, alleviate the workload

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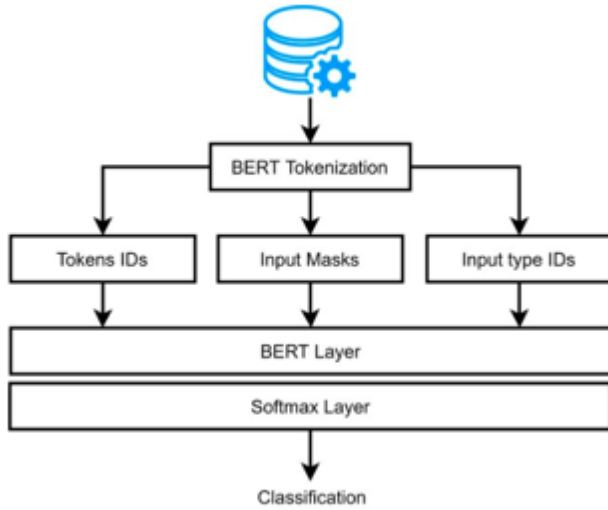


Figure 1: Classifications steps in BERT

of legal practitioners, and offer a cost-effective alternative to traditional legal consultation methods. In the context of Moroccan labor law, a chatbot can serve as a valuable initial point of reference, providing preliminary advice and information that would otherwise require extensive and potentially expensive legal counsel. The integration of chatbots into legal practices has become an increasingly intriguing topic in the field of legal technology. Research indicates that legal chatbots can effectively assist users in understanding complex legal terminologies and processes [1]. For example, Bayamlıoglu and Leenes (2020) [1] discuss the role of chatbots in democratizing access to legal information. Similarly, McGinnis and Pearce (2019) [2] explore the transformative potential of AI in legal services, highlighting how technologies like chatbots are reshaping the landscape of legal advice. These scholarly works provide a foundational understanding of the impact and potential of a labor law chatbot within the Moroccan context.

2.2 Overview of Current Technologies in Legal Chatbots

Recent advancements in Artificial Intelligence, particularly in the fields of Natural Language Processing (NLP) and Machine Learning (ML), have had a profound impact on the capabilities of legal chatbots. Notably, technologies such as BERT [5] and GPT [6] have played a crucial role in enhancing the text processing and generation capabilities of these chatbots, enabling them to handle complex legal language more effectively (as illustrated in Figure 1).

The integration of chatbots in the field of law highlights their immense potential in delivering accessible legal guidance. In the case of Moroccan labor law, which is known for its intricacies, these technologies can greatly streamline the process of legal consultation. A notable advancement in chatbot technology is the Retrieval-Augmented Generation (RAG) model [7]. Unlike models that solely rely on pre-trained datasets, RAG combines information retrieval with advanced language generation techniques, resulting in more

accurate and contextually relevant responses. This development represents a significant leap forward in the capabilities of chatbots.

3 OUR PROPOSED APPROACH: THE RAG METHOD

3.1 Description of the RAG model

The Retrieval-Augmented Generation (RAG) model is a pivotal development in artificial intelligence for natural language understanding. It combines two key processes:

Information Retrieval: The model retrieves relevant information using a query encoder ($q = \text{Eq}(Q)$) and a document decoder ($D = \text{Ed}(\text{Doc})$). Here, Q represents the user query, Eq the query encoder, Doc the document from the database, and Ed the document decoder. This process can be further represented by the retrieval function $R(q)$, which fetches the most relevant document for the given query [7].

Language Generation: The model generates responses based on the retrieved information, using a generative language model. This is mathematically represented as $P(\text{Response}|Q, D) = \prod_i P(w_i|w_{<i}, Q, D)$, indicating the probability of generating a response word by word, given the query and retrieved document [7].

3.2 Advantages of RAG Over Other AI Models

One of the key advantages of the Retrieval-Augmented Generation (RAG) model is its ability to incorporate real-time information from external sources. This dynamic integration is represented by the equation $R = G(q, D)$, where R denotes the generated response and G represents the generation model. This feature is particularly valuable in areas such as labor law, where regulations are subject to constant evolution [8]. By integrating RAG into chatbots, the quality of interaction is significantly enhanced. In the context of law chatbots, RAG enables the access of up-to-date legal information and the generation of personalized responses. The overall process can be summarized as $\text{Response} = \text{RAG}(Q)$, encompassing query interpretation, information retrieval, and response generation [9].

3.3 RAG vs Fine-tuning

Fine-tuning involves adjusting the parameters θ of a pre-trained model M on a specific dataset D . The optimization is represented as:

$$\theta^* = \arg \min_{\theta} L(M(D; \theta), Y)$$

where L represents the loss function measuring the difference between the model's output $M(D; \theta)$ and the true values Y [10]. The goal is to find the optimal parameters θ^* that minimize this loss.

The RAG model combines retrieval and generation [7]. For a query Q , the retrieval function selects relevant documents D_{ret} from a database, and the generation uses these documents to construct a response R . This process is represented as:

Retrieval: $D_{\text{ret}} = \text{Retrieve}(Q, D)$ $\text{Retrieve}(\cdot)$ maps the query Q to a set of documents D_{ret} from the database D .

Generation: $R = \text{Generate}(Q, D_{\text{ret}}; \theta)$ $\text{Generate}(\cdot)$ is the generative model producing the response R using both the query Q and retrieved documents D_{ret} , parameterized by θ .

3.4 The motivation of choosing RAG for This Project

For our system, the RAG model was chosen over fine-tuning for several key reasons:

- **Dynamic Content Integration:** RAG dynamically integrates real-time data, making it suitable for rapidly changing information areas. This integration is mathematically represented by the function $R = G(\text{Retrieve}(Q), \theta)$ [7], where $\text{Retrieve}(Q)$ fetches the most recent data relevant to query Q , and G generates the response using parameters θ .
- **Contextual Relevance:** RAG's retrieval mechanism ensures contextually relevant responses by leveraging recent and pertinent legal information [7]. The relevance can be quantified using a similarity measure $\text{sim}(Q, \text{Dret})$, where higher values indicate greater relevance of the retrieved documents Dret to the query Q .
- **Scalability:** RAG allows for scalability in data by accessing an ever-expanding knowledge base [7]. This is unlike static training datasets and can be represented by the model's ability to handle an increasing volume of data.

So, the RAG approach offers several advantages in the context of law chatbot:

- **Up-to-Date Legal Information:** RAG's retrieval aspect ensures access to current legal information, essential for accurate legal advice [7]. This feature is critical in a domain where laws and regulations are frequently updated.
- **Customized Responses:** The generative component of RAG allows for tailored responses to user-specific queries, enhancing engagement [7].
- **Reduced Training Data Dependency:** RAG reduces dependency on extensive labeled training datasets, beneficial in specialized fields like law [7]. This reduction in dependency is represented by the model's ability to utilize external databases effectively, reducing the need for large-scale, labeled datasets. For further understanding of these concepts, refer to the foundational work on RAG and NLP models [5, 7].

4 EXPERIMENTATIONS AND DISCUSSION

4.1 The development of chatbot

During the experimentation process of our proposed approach, several key stages were undertaken:

- **Data Collection on Law field:** This phase involved the comprehensive collection of Moroccan labor law data, including articles from the Moroccan Labor Code and the Dahirs (royal decrees). It was crucial to gather all relevant legal texts, amendments, and updates to ensure a complete representation of the current legal framework [4].
- **Processing and Segmenting Labor Law PDFs:** This stage focused on segmenting the collected legal documents into smaller, manageable chunks. The objective was to determine the optimal chunk size for efficient data retrieval. This step was vital in processing complex legal texts and making them more accessible for analysis.

- **Embedding with a Hugging Face Model:** A Hugging Face model specifically designed for text processing was utilized in this phase. The segmented legal documents were embedded into a machine-readable format, enabling advanced natural language processing for the extraction and analysis of legal language.
- **Creation of an SQL Database:** This step involved structuring the processed data into an SQL database. The database stored vectors representing the segmented chunks of text, enabling efficient data retrieval and processing. This approach facilitated quick and accurate access to specific information within the extensive corpus of legal text.
- **Formulating Chatbot Query Templates:** This stage focused on designing query templates to facilitate effective user interactions with the chatbot. These templates guided users in framing their inquiries in a comprehensible manner for the chatbot, ensuring accurate responses.
- **Using the Llama-70B Model for the Chatbot:** The Llama-70B model was integrated into the chatbot for advanced language processing. This model enhanced the chatbot's ability to understand complex legal language and provide precise answers to user queries [11].
- **Implementation in a User Interface with Angular and Django:** The chatbot was integrated into a user interface developed using Angular and Django. Angular was utilized for building a dynamic front-end, while Django managed backend interactions, ensuring a seamless user experience.

4.2 The integration of RAG approach in chatbot

The integration of the Retrieval-Augmented Generation (RAG) model into the chatbot's architecture is a crucial enhancement that greatly enhances its capabilities in processing and responding to user queries about labor law. This integration encompasses several key aspects:

- **Query Processing and Document Retrieval:** When a user enters a query, the RAG method first processes the query to understand its context and intent. It then retrieves the most relevant documents or data segments from the SQL database, which contains embedded vectors representing legal texts. The retrieval process is based on the semantic similarity between the query and the contents of the database, ensuring that the most pertinent information is selected.
- **Response Generation:** Once the relevant data is retrieved, the chatbot utilizes the LLama 70-B model, a variant of large language models, to generate the response. This model leverages the retrieved information to construct a coherent and contextually appropriate answer. Unlike simply regurgitating database contents, the LLama 70-B synthesizes information from multiple sources, utilizing its advanced language processing capabilities to generate natural-sounding and informative replies. This integration ensures that the chatbot's responses are not only accurate but also finely tuned to the nuances of the user's query.
- **Continuous Learning and Updating:** A key feature of the RAG model in our chatbot is its ability to continuously learn and update its responses. As new legal information is added

Table 1: Overall performance on grouped academic benchmarks compared to open-source base models.

Model and Size	Code Reasoning	AGI EVAL
MPT-7B	20.5	23.5
MPT-30B	28.9	33.8
Falcon-7B	5.6	21.8
Falcon-40B	15.2	37.0
Llama1-7B	14.1	23.9
Llama1-65B	30.7	47.6
Llama2-7B	16.8	29.3
Llama2-34B	27.8	43.4
Llama2-70B	37.5	54.2

to the database, the chatbot becomes more proficient in handling diverse and complex queries, ensuring that users always receive the most up-to-date and accurate legal advice.

- **Integration with User Interface:** The final aspect of RAG's integration is its seamless operation within the chatbot's user interface, which is developed using Angular and Django. This ensures that users interact with the chatbot in an intuitive and user-friendly environment, where the advanced capabilities of the RAG model are effectively utilized to enhance the user experience. This sophisticated integration of the RAG method into the chatbot's structure is pivotal in achieving high accuracy, relevance, and user satisfaction in the domain of Moroccan law consultation.

Thus, the integration of LLama 70-B significantly enhances the chatbot's ability to provide precise and context-aware legal advice, solidifying its position as a cutting-edge tool in AI-driven legal assistance [11].

4.3 Discussion of results.

The integration of the Retrieval-Augmented Generation (RAG) method into the development of our law chatbot represents a major advancement in the application of artificial intelligence to legal assistance. This innovative approach stands out as a pioneering solution in the field of legal technology, particularly in the context of Moroccan labor law. By combining efficient information retrieval with advanced natural language generation, the RAG method empowers the chatbot to deliver precise and contextually relevant legal information. This is particularly important given the complexity and nuances of labor laws, which often involve intricate terminology and legal principles.

The chatbot's ability to interpret and respond to user queries with accurate legal information enhances the user experience and promotes the democratization of legal knowledge. It breaks down language barriers and simplifies legal jargon, making it an invaluable tool for individuals and professionals who may lack immediate access to legal expertise or resources. Moreover, the successful integration of RAG into a user-friendly chatbot interface highlights the potential of AI to revolutionize legal consulting services. This technology goes beyond traditional text-based legal information systems, offering interactive and personalized legal guidance.

The design of the chatbot prioritizes ease of use and accessibility, ensuring that users from diverse backgrounds, regardless of their familiarity with legal concepts, can benefit from this service. It addresses a significant need in the legal domain – the simplification and clarification of legal procedures and regulations for the general public. This advancement not only demonstrates the capabilities of AI in enhancing service delivery but also sets a precedent for future applications of AI in legal technology. The development of this chatbot could pave the way for similar initiatives in other legal systems and jurisdictions, potentially transforming how legal information is disseminated and accessed globally.

The deployment of RAG within the chatbot framework marks a substantial improvement over traditional chatbot technologies. The model's ability to dynamically integrate real-time legal data and generate contextually relevant responses enriches the user experience and ensures the accuracy and timeliness of the legal guidance provided. This has significant implications, especially in a legal landscape that is constantly evolving. Users can rely on the chatbot for immediate and reliable legal advice, which is particularly beneficial for those who may not have easy access to legal experts.

5 CONCLUSION AND PERSPECTIVES

The development of this law chatbot using the RAG model is a significant contribution to the field of legal technology. It showcases the immense potential of AI in revolutionizing access to legal information, making it more readily available, accurate, and understandable to a broader audience. As the realms of AI and legal tech continue to progress, projects like this one will play a pivotal role in shaping the future of legal consultation and services.

From a forward-looking perspective, this project opens up several avenues for further research and development:

- **Expansion to Other Legal Domains:** While the current chatbot is specifically tailored to Moroccan labor law, the underlying technology has the potential to be adapted to other legal fields and jurisdictions. This expansion would greatly broaden the impact and usefulness of the chatbot technology in making legal advice more accessible across various legal domains.
- **Enhanced Natural Language Understanding:** Future iterations of the chatbot could focus on improving its natural language processing capabilities. This would enable the

chatbot to better understand the nuances of user queries and engage in more sophisticated and meaningful interactions.

- **Continuous Learning and Updating:** Implementing mechanisms for continuous learning would empower the chatbot to autonomously adapt to legal changes over time. This would reduce the need for manual updates and ensure that the chatbot remains relevant and accurate in the face of evolving legal landscapes.

By exploring these areas of further research and development, we can continue to refine and advance the capabilities of the chatbot, making it an even more valuable tool for legal assistance and consultation.

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